

Yiqi Zhao

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EDUCATION

University of Southern California

PhD Candidate in Computer Science

Los Angeles, USA

2023/08 - Present

- GPA: 4.0/4.0
- Supervisors: [Prof. Jyotirmoy V. Deshmukh](#); [Prof. Yue Wang](#)
- Viterbi School of Engineering Fellowship

University of Southern California

Master of Science in Computer Science

Los Angeles, USA

2023/08 - 2025/08

- GPA: 4.0/4.0
- Petitioned during PhD study.

Vanderbilt University

Bachelor of Science, Magna Cum Laude

Nashville, USA

2020/08 - 2023/05

- GPA: 3.935/4.0
- Majors: Honors in Computer Science; Mathematics
- Minors: Electrical Engineering; Data Science
- Dean's List; Good Standing

WORK EXPERIENCE

Research and Development Intern, Toyota Research Institute of North America

Supervisors: [Georgios Fainekos](#), [Bardh Hoxha](#)

Ann Arbor, USA

2025/05 - 2025/08

- Worked on Cyber-Physical Systems research at the Future Research Department.
- Focused on Model Predictive Path Integral for Path Planning.

Application Development Intern, ADP

Supervisor: [Jim Farber](#)

Roseland, USA

2022/06 - 2022/08

- Full stack API development for the Autopay Group within the Global Product & Tech group.

RESEARCH INTERESTS

Robotics, Cyber-Physical Systems, Formal Methods, Systems and Control Theory

PREPRINTS (* INDICATES EQUAL CONTRIBUTION; † DENOTES EQUAL ADVISING)

[p6] Kehan Long, **Yiqi Zhao**, Pol Mestres, Lars Lindemann, Nikolay Atanasov, Jorge Cortés. "A Unified Perspective on Conformal Prediction and Wasserstein Distributionally Robust Optimization for Uncertainty Quantification". *Arxiv*, 2026. *Currently under review in Annual Reviews in Control*. [\[paper\]](#)

[p5] **Yiqi Zhao**, Taekyung Kim, Hideki Okamoto, Bardh Hoxha, Jyotirmoy V. Deshmukh, Lars Lindemann, Georgios Fainekos. "Safety-aware Model Predictive Path Integral Control with Signal Temporal Logic". *Arxiv*, 2026. [\[paper\]](#)

[p4] Celina Shiyu Wang, **Yiqi Zhao**[†], Junjie Ye, Yue Wang, Jyotirmoy V. Deshmukh[†]. "Logic-VLA: A Temporal Logic Conditioned Vision-Language-Action Model". *IROS 2026 Workshop on Learning and Formal Methods for Robotics*. [\[paper\]](#)

[p3] **Yiqi Zhao**^{*}, Ruohai Ge^{*}, Celina Shiyu Wang, Junjie Ye, Muchen Xu, Minhao Li, Sergey Zakharov, Basile Van Hoorick, Vitor Campagnolo Guizilini, Leonidas Guibas, Gaurav S. Sukhatme, Jyotirmoy V. Deshmukh, Yue Wang. "DUET: Dual-Robot Understanding via Efficient Teaching". *Best Paper Finalist at the IROS 2026 Workshop on Physical Collaboration and Interaction in Multi-Robot Systems*. [\[paper\]](#) [\[project\]](#)

[p2] **Yiqi Zhao**^{*}, Xinyi Yu^{*}, Matteo Sesia, Jyotirmoy V. Deshmukh, and Lars Lindemann. "Conformal Predictive Programming for Chance Constrained Optimization". *Arxiv*, 2025. *Currently under review in Automatica*. [\[paper\]](#) [\[codes\]](#)

[p1] **Yiqi Zhao**, Ziyan An, Meiyi Ma, and Taylor Johnson. "EduSAT: A Pedagogical Tool for Theory and Applications of Boolean Satisfiability". *Arxiv*, 2023. [\[paper\]](#) [\[tool\]](#)

CONFERENCE PAPERS

[c3] Anand Balakrishnan, Rohit Bernard, Shreeram Narayanan, Vidisha Kudalkar, **Yiqi Zhao**, Parinitha Nagaraja, Georgi Markov, Christof Budnik, Helmut Degen, Lars Lindemann, Jyotirmoy V. Deshmukh. "Safety Assurance for Autonomous Systems with Multiple Sensor Modalities". *The 22nd ACM-IEEE International Symposium on Formal Methods and Models for System Design (MEMOCODE)*, Raleigh, USA. 2024. [\[paper\]](#)

[c2] **Yiqi Zhao**, Bardh Hoxha, Georgios Fainekos, Jyotirmoy V. Deshmukh, and Lars Lindemann. "Robust Conformal Prediction for STL Runtime Verification under Distribution Shift". *The 15th ACM/IEEE International Conference on Cyber-Physical Systems*, Hong Kong, China. 2024. **Best Paper Award Finalist**. [\[paper\]](#) [\[codes\]](#)

[c1] **Yiqi Zhao**, Ziyang An, Xuqing Gao, Ayan Mukhopadhyay, Meiyi Ma. "Fairguard: Harness Logic-based Fairness Rules in Smart Cities". *The 8th ACM/IEEE Conference on Internet of Things Design and Implementation*, San Antonio, USA. 2023. [\[paper\]](#)

JOURNAL PAPERS (* INDICATES EQUAL CONTRIBUTION.)

[j5] Lars Lindemann, **Yiqi Zhao**, Xinyi Yu, George J. Pappas, Jyotirmoy V. Deshmukh. "Formal Verification and Control with Conformal Prediction: Practical Safety Guarantees for Autonomous Systems". *IEEE Control Systems*. 2025. [\[paper\]](#) [\[tutorial\]](#)

[j4] **Yiqi Zhao***, Xinyi Yu*, Bardh Hoxha, Georgios Fainekos, Jyotirmoy V. Deshmukh, and Lars Lindemann. "STL-GO: Spatio-Temporal Logic with Graph Operators for Distributed Systems with Multiple Network Topologies". *ACM Transactions on Embedded Computing Systems; the ACM SIGBED International Conference on Embedded Software (EMSOFT 2025, ESWEEK)*, Taipei, 2025. [\[paper\]](#)

[j3] Xinyi Yu, **Yiqi Zhao**, Xiang Yin, and Lars Lindemann. "Signal Temporal Logic Control Synthesis among Uncontrollable Dynamic Agents with Conformal Prediction". *Automatica*. 2025. [\[paper\]](#) [\[codes\]](#) (**Regular Paper.**)

[j2] **Yiqi Zhao**, Emily Zhu, Bardh Hoxha, Georgios Fainekos, Jyotirmoy V. Deshmukh, Lars Lindemann. "Distributionally Robust Predictive Runtime Verification under Spatio-Temporal Logic Specifications". *ACM Transactions on Cyber-physical Systems*. 2025. [\[paper\]](#) [\[codes\]](#)

[j1] Ziyang An, **Yiqi Zhao**, Xuqing Gao, Ayan Mukhopadhyay, Meiyi Ma. "Formal Logic-Guided Harnessing Heterogeneous Fairness Rules in Smart Cities". *ACM Transactions on Cyber-physical Systems*. 2025. [\[paper\]](#)

NON-AUTHORSHIP ACKNOWLEDGED CONTRIBUTIONS

[o2] Junyang Cai, Weimin Huang, Jyotirmoy V. Deshmukh, Lars Lindemann, Bistra Dilkina. "Neuro-Symbolic Acceleration of MILP Motion Planning with Temporal Logic and Chance Constraints". *Arxiv*, 2025. [\[paper\]](#) (Acknowledged for the help.)

[o1] Hanwen Liu, Shashank Giridhara, Ibrahim Sabek. "Conformal Prediction for Verifiable Learned Query Optimization". *Proceedings of the VLDB Endowment*. 2025. [\[paper\]](#) (Acknowledged for the support.)

SELECTED HONORS

Best Paper Finalist at the IROS 2026 Workshop on Physical Collaboration and Interaction in Multi-Robot Systems Sep. 2026
Awarded at the IROS 2026 Workshop on Physical Collaboration and Interaction in Multi-Robot Systems for the paper "DUET: Dual-Robot Understanding via Efficient Teaching".

ICCPS Best Paper Award Finalist May. 2024
Awarded at the 15th ACM/IEEE International Conference on Cyber-Physical Systems for the paper "Robust Conformal Prediction for STL Runtime Verification under Distribution Shift".

Viterbi School of Engineering Fellowship Aug. 2023
Awarded for selected incoming PhD students at Viterbi School of Engineering, USC.

WISE Award (Vanderbilt Institute for Surgery and Engineering Summer Fellowship) 2021
Awarded for the paid fellowship at the Vanderbilt Institute for Surgery and Engineering Summer Fellows Program.

TEACHING EXPERIENCE

Teaching Assistant of CSCI 513 (Autonomous Cyber-Physical Systems), University of Southern California Los Angeles, USA
Supervisors: [Jyotirmoy V. Deshmukh](#) 2026/09

- Grading students' homework, miniprojects, and holding office hours.

Teaching Assistant of CSCI 577A (Software Engineering), University of Southern California Angeles, USA <i>Supervisors: Jae Young Bang, Mahdi Eslamimehr</i> • Graded students' homework and exams.	Los Angeles, USA 2026/01 - 2026/05
Grader of CS 4260 (Artificial Intelligence), Vanderbilt University <i>Supervisors: Meiyi Ma, Daniel Moyer</i> • Graded students' homework and exams. • Held TA office hours regularly.	Nashville, USA 2022/08 - 2023/05

ACADEMIC SERVICES

Journal Reviews for NAHS

Conference Reviews for ACC 2024, VMCAI 2024, CDC 2024, ICLR 2025, AISTATS 2025, HSCC 2025, L4DC 2025, ICCPS 2025, NEUS 2025, MECC 2025, ICRA 2026, AISTATS 2026, L4DC 2026, NeurIPS 2026.

Conference PC Member and Reviews for HSCC 2024 (Repeatability Evaluation), HSCC 2025 (Repeatability Evaluation), ICCPS 2025 (Posters and Demos).

Tutorial on Formal Verification and Control with Conformal Prediction at the CPS-IoT Week 2025 [[home-page](#)].
05/06/2025

SELECTED TALKS

The 15th ACM/IEEE International Conference on Cyber-Physical Systems	05/16/2024
The 43rd Southern California Control Workshop (with Xinyi Yu)	04/19/2024
The 8th ACM/IEEE Conference on Internet of Things Design and Implementation	May 2023

SKILLS

Programming Languages	Python, Javascript, C++, C, MATLAB, Java, SQL. Mandarin Chinese (native), English (TOEFL: 116).
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